Seifeddine Reguige

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Education

University of Ottawa Ottawa, ON
Bachelor of Science, Computer Science *Graduated Aug. 2026*

- CGPA: 8.18/10 | Dean's Honour List (3x: Fall 2023, Fall 2024, Winter 2026) | 2x Recognition of Excellence award
- Relevant coursework: Design & Analysis of Algorithms, Data Structures, Object-Oriented Programming (C++), Operating Systems, Computer Networks & Protocols, Secure Systems Design, Databases, Artificial Intelligence, Fundamentals of Data Science, Software Requirements Analysis, Design & Analysis of User Interfaces

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, PHP, GoLang, Prolog, Scheme, HTML/CSS, Bash
Frontend, Mobile & UI/UX: React Native, Expo (Expo Router), Android Studio, responsive & accessible UI development, wireframing & prototyping, usability heuristics, user-centered design principles
Backend & Databases: REST APIs, Supabase (Postgres, RLS, Auth, Realtime, Storage), Firebase, SQLite, relational data modeling, ETL pipelines
ML & Data Science: Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, OpenCV, XGBoost, statistical modeling, regression, clustering, recommender systems, feature engineering, data visualization
Tools & Practices: Git, GitHub Actions (CI/CD), Docker, Kubernetes, Streamlit, Jupyter, VS Code, Linux/Unix, Agile/Scrum, technical documentation

Experience

Research Assistant / Data Scientist Summer 2026 – Present
University of Ottawa – Faculty of Social Sciences *Ottawa, ON*

- Contribute to a secure digital research infrastructure for sensitive public health and policy research data, applying data governance and controlled access principles across the pipeline
- Apply core data science practices – data cleaning, normalization, exploratory data analysis, metadata documentation, and quality validation – to migrate legacy datasets into analysis-ready form
- Collaborate with researchers and infrastructure teams to define data models, document variables, and design future relational database integration

IT Systems Developer Oct. 2024 – Apr. 2025
Faculty of Health Sciences, University of Ottawa *Ottawa, ON*

- Developed and maintained web-based systems and data-driven interfaces using PHP for faculty platforms
- Worked with backend logic, API interactions, and structured data processing to support research and operational workflows
- Built responsive interfaces and improved usability for faculty members, staff, and researchers

Projects

Metadata-Based Optical Flow Reliability Predictor | *Python, PyTorch, OpenCV, scikit-learn, Streamlit* 2026

- Built an end-to-end ML pipeline to predict optical-flow reliability from scenario metadata using TFRecord-based video datasets and engineered motion features
- Evaluated Farneback and RAFT error labels using Endpoint Error, then compared Linear Regression, Random Forest, and Gradient Boosting under scenario-holdout validation
- Created a Streamlit explorer with flow visualizations, EPE heatmaps, and qualitative analysis tools for model interpretation

property-ops – Role-Based Property Management Platform | *React Native, Expo, TypeScript, Supabase* 2026

- Building a cross-platform short-term rental management app for hosts, co-hosts, cleaners, and maintenance staff with role-based access control via Postgres RLS
- Implementing compliance tracking, revenue dashboards, shift/turnover management, and maintenance issue tracking with Supabase Realtime
- Following professional engineering practices: branch-protected main, feature-branch PRs, GitHub Actions CI (lint + typecheck), Conventional Commits

PCSurMesure – Role-Based Mobile Ordering System | *Java, Android Studio, Supabase, REST APIs* 2025

- Led development of a multi-role Android ordering application with authentication, permission-controlled workflows, and order lifecycle management
- Migrated local SQLite storage to Supabase and applied MVC architecture to improve scalability and maintainability

Movie Recommender System | *Python, Pandas, NumPy, scikit-learn, Surprise* 2025

- Built and compared content-based filtering, clustering, and collaborative filtering approaches on The Movies Dataset
- Engineered sparse utility matrices for memory-efficient matrix factorization; evaluated relevance using SVD, MSE, Precision@K, and Mean Reciprocal Rank